Prepared for submission to JHEP

Eikonal Time Delay and Post-Newtonian Constraints in Bimetric Gravity

Mohamed Amine Ezzine

The Royal Institute of Technology, KTH, Stockholm, Sweden

E-mail: ezzine@kth.se

Abstract: We compute the gravitational time delay in ghost-free bimetric gravity using an amplitude-based eikonal approach. By modeling the interaction between a photon and a massive source through graviton exchange, we derive the eikonal phase from the scattering amplitude and extract the corresponding classical time delay. In the weak-field regime, the result naturally separates into contributions from the massless and massive spin-2 modes, with the former reproducing the Shapiro time delay of general relativity. The massive mode induces a correction to the effective post-Newtonian (PPN) parameter γ, allowing for a direct comparison with Solar System observations. Using constraints from the Cassini mission, we derive bounds on the parameter space of the theory in terms of the coupling strength and the Fierz–Pauli mass. Our results provide an amplitude-based derivation of classical gravitational observables in bimetric gravity and establish consistency with existing field-theoretic approaches in the appropriate limits.

Contents

1 Introduction **1**

2 Scalar Vector Interaction in Bimetric Theory **3**
- 2.1 Bimetric Gravity and Proportional Solutions 3
- 2.2 Mass Eigenstates and Linearized Action 4
- 2.3 Vector Scalar Gravitational Interaction 5

3 Eikonal Scattering Matrix and Time Delay Computation **8**
- 3.1 Kinematic Notations and Fourier Transforms Definitions 8
- 3.2 Tree Level Amplitudes and Vertices Simplifications 10
- 3.3 Eikonal Phase and Time Delay Computation 11

4 PN Parameter Computation and Parameter's Constraints From Solar System Measurements **14**
- 4.1 Frequency Shift Computation 14
- 4.2 Post-Newtonian Parameter Derivation from Time Delay Measurements 15

5 Conclusion **17**

A Derivation of the General Ladder Diagram Contribution **17**

B Shapiro Time Delay from Massless mode contribution **21**

1 Introduction

Bimetric Theory (BT) describes the non-derivative interaction of two spin-two fields. It can be seen as an extension of Einstein's theory of General relativity (GR) where gravitational interactions are encoded in one massless spin-two field representing the metric $g_{\mu\nu}$. Generically, theories involving two interacting metrics suffer from the Boulware-Deser ghost, a propagating degree of freedom with negative kinetic energy [1]. However, the special structure of the interaction between the metrics in BT ensures the absence of such pathological degrees of freedom.

As a gravitational theory BT has a rich phenomenology. The existence of the second metric modifies the Friedmann equations. It was shown that these new modified equations have expanding solutions without invoking a cosmological constant [2], thus providing an explanation for the problem of dark energy. Furthermore, while an invariant decomposition does not exist in general cases, it is possible for specific settings to decompose BT into a massive and massless spin-two mode. The massive mode can be a candidate for dark matter [3].

The free parameters of the theory are given by

- m_g, m_f the Planck masses for each metric, specifying the interaction scale with matter fields. We define the parameter $\alpha = \frac{m_g}{m_f}$.
- $\beta_1 ... \beta_4$ which encode the strength of each term in the interaction potential of two metrics.

Given its rich phenomenology, there has been a significant effort in trying to set observational constraints on these parameters. Three main approaches have been employed:

1. Cosmological observations : The modification of the cosmological equations due to the presence of the second metric changes the expansion and structure formation profile, resulting in potential deviations from the standard cosmological model, ΛCDM. Varying the parameters β_i in the interaction potential of the theory gives rise to different cosmological models. In [4, 5], the authors used the data from type Ia supernovae to derive restrictions on the different parameters to match the expansion history of the universe.

2. Gravitational waves measurements: As mentioned earlier, it is possible to decompose perturbations around certain backgrounds into a massive and massless gravitons. Coherent states of these perturbations constitute gravitational waves. In GR such perturbations are massless and therefore propagate at the speed of light. The presence of a massive mode then influence the propagation speed. In [6], by comparing the arrival time of electromagnetic signals with that of gravitational signals, an upper bound on the mass of the massive gravitons was derived. [7] considers the possible constraint on BT from a possible future measurements of primordial gravitational waves, it was shown that BT is subject to less sever constraints than massive gravity.

3. Solar System Measurements: Since its proposal in 1964 [8] and its subsequent measurement in 2003 during the Cassini space mission [9], Shapiro time delay in the Solar System have provided the strongest constraints on any deviation from GR. In reference [10] the Post Newtonian (PPN) parameter for BT was computed. By first performing a perturbative expansion of the theory then solving the resulting equations, constraints on the value of α and the mass of the massive graviton field were derived.

In this paper we provide an alternative computation of the time delay, and hence the PPN parameter, using Quantum Field Theory (QFT) techniques. We model the propagation of light in the Solar field through the gravitational interaction of the photon field A_μ with a massive scalar field ϕ representing the Sun. In the weak-field limit this interaction is mediated by the exchange of both massive and massless gravitons. In the eikonal approximation the center of mass energy s is leading compared to the exchanged momenta q. This is the appropriate regime to describe light propagation in the vicinity of the sun [11, 12]. As we will prove later, in the eikonal regime, the scattering matrix takes an exponential form simplifying the computation of time delay.

While the eikonal approach to classical gravitational observables is well established in GR [13] and its higher-curvature extensions [14], its application to theories with multiple interacting spin-2 fields presents new challenges and new physics. In BT, the presence of a massive graviton mode alongside the massless one modifies the eikonal phase in a way that is directly tied to the PPN parameter γ. This work provides the first amplitude-based derivation of this correction, establishing a transparent connection between the microscopic graviton exchange and the macroscopic time delay measured in Solar System experiments such as the Cassini mission [9].

The rest of this paper is structured as follows: In section 2, we start with a short summary of BT, after which we derive the Feynman rules for the interacting system. In section 3, we compute the scattering matrix elements in the eikonal regime and derive the time delay. In section 4 we calculate the PPN and use the Cassini measurements to derive constraints on BT. We conclude this section by comparing our results with those of [10].

2 Scalar Vector Interaction in Bimetric Theory

2.1 Bimetric Gravity and Proportional Solutions

In the absence of matter coupling, the action of BT is given by [15]

$$S_{BT} = m_g^2 \int d^4x\sqrt{-g}\,\mathcal{R}(g) + m_f^2 \int d^4x\sqrt{-f}\mathcal{R}(f) - 2m^4 \int d^4x\sqrt{-g}\sum_{n=1}^{4}\beta_n e_n(S). \quad (2.1)$$

- $g_{\mu\nu}$ and $f_{\mu\nu}$ are the two metrics representing the spin-two fields. Throughout this work, we assume that they have a mostly positive signature.

- $\mathcal{R}(g)$ and $\mathcal{R}(f)$ are the Ricci scalars of the metrics $g_{\mu\nu}$ and $f_{\mu\nu}$.

- g and f are the determinants of the metrics.

- m_g and m_f are the Planck masses.

- S is the matrix defined by $(S^2)^\mu_\nu = g^{\mu\rho}f_{\rho\nu}$.

- $e_n(S)$ are the elementary symmetric polynomials defined recursively by

$$e_0(S) = 1 \quad (2.2)$$
$$e_n(S) = \frac{(-1)^{n+1}}{n}\sum_{k=0}^{n-1}(-1)^k[S^{n-k}]e_k(S). \quad (2.3)$$

  $[S]$ is the trace of the matrix S.

- β_i are dimensionless parameters characterizing the relative strength of each interaction term.

- m is a global parameter characterizing the overall strength of the interaction.

The dynamics of the two metrics $g_{\mu\nu}$ and $f_{\mu\nu}$ are governed by the following coupled equations of motion

$$\mathcal{R}_{\mu\nu}(g) - \frac{1}{2}g_{\mu\nu}\mathcal{R}(g) + \frac{m^4}{m_g^2}V^g_{\mu\nu}(g, f, \beta_i) = 0, \tag{2.4a}$$

$$\mathcal{R}_{\mu\nu}(f) - \frac{1}{2}f_{\mu\nu}\mathcal{R}(f) + \frac{m^4}{m_f^2}V^f_{\mu\nu}(g, f, \beta_i) = 0. \tag{2.4b}$$

The exact expression for the potentials $V^g_{\mu\nu}$ and $V^f_{\mu\nu}$ can be found in [16]. For our present purpose, what we need is the existence of proportional solutions [17]. This is a class of solutions for which the two metrics obey the proportionality relation

$$f_{\mu\nu} = c^2 g_{\mu\nu}. \tag{2.5}$$

In this case the equations of motion simplify to two copies of Einstein's equations:

$$\mathcal{G}_{\mu\nu}(g) + \Lambda_g(c, \alpha, \beta_n)g_{\mu\nu} = 0, \tag{2.6}$$

$$\mathcal{G}_{\mu\nu}(f) + \frac{1}{c^2}\Lambda_f(c, \alpha, \beta_n)f_{\mu\nu} = 0. \tag{2.7}$$

Given that the Einstein Tensor $\mathcal{G}$ is scale invariant, the two previous equations result in the following conditions

$$\Lambda_g(c, \alpha, \beta_n) = \Lambda_f(c, \alpha, \beta_n). \tag{2.8}$$

This in turn fixes the value of c in terms of the rest of the theory parameters.

The importance of the proportional solutions is twofold: (i) they reproduce all GR solutions. (ii) They play an important role as background on top of which perturbations of the metric can be diagonalized in an invariant manner into those of massive and massless spin-two fields. The second point is of high importance, since this decomposition will allow us to distinguish the contribution to the time delay from the massless GR-like mode and the massive mode.

2.2 Mass Eigenstates and Linearized Action

Let $\bar{g}_{\mu\nu}$ and $\bar{f}_{\mu\nu} = c^2\bar{g}_{\mu\nu}$ be two proportional solutions for the bimetric equations. We consider the following perturbations

$$f_{\mu\nu} = \bar{f}_{\mu\nu} + \frac{1}{m_f}\delta f_{\mu\nu}, \tag{2.9}$$

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \frac{1}{m_g}\delta g_{\mu\nu} \tag{2.10}$$

The mass eigenstates are expressed in terms of the perturbations $\delta g_{\mu\nu}$ and $\delta f_{\mu\nu}$ as

$$G_{\mu\nu} = \delta f_{\mu\nu} + \alpha^2\delta g_{\mu\nu}, \tag{2.11}$$

$$M_{\mu\nu} = \frac{1}{2c}\left(\delta f_{\mu\nu} - \delta g_{\mu\nu}\right). \tag{2.12}$$

$G_{\mu\nu}$ and $M_{\mu\nu}$ correspond respectively to the massless and massive graviton fields. $G_{\mu\nu}$ obeys the linearized Einstein's equation

$$\tilde{\varepsilon}_{\mu\nu}{}^{\rho\sigma} G_{\rho\sigma} - \Lambda_g \left(G_{\mu\nu} - \frac{1}{2}\tilde{g}_{\mu\nu} G \right) = 0, \tag{2.13}$$

while $M_{\mu\nu}$ obeys the Fierz–Pauli massive spin-two equation [18]

$$\begin{aligned}\tilde{\varepsilon}_{\mu\nu}{}^{\rho\sigma} M_{\rho\sigma} - & \Lambda_g \left(M_{\mu\nu} - \frac{1}{2}\tilde{g}_{\mu\nu} M \right) \\ & + \frac{m_{FP}^2}{2} \left(M_{\mu\nu} - \tilde{g}_{\mu\nu} M \right) = 0.\end{aligned} \tag{2.14}$$

$\tilde{g}_{\mu\nu} = (1+\alpha^2)\bar{g}_{\mu\nu}$ is the rescaled background metric. $G = \tilde{g}^{\mu\nu} G_{\mu\nu}$, and M is defined analogously. m_{FP} is the Fierz–Pauli mass and it is given in terms of the theory parameters by

$$m_{FP}^2 = \frac{m^4}{m_g^2} \frac{1}{\alpha^2 c^2} (c\beta_1 + 2c^2\beta_2^2 + c^3\beta_3) \tag{2.15}$$

The Kinetic operator $\tilde{\varepsilon}_{\mu\nu}{}^{\rho\sigma}$ is given by

$$\tilde{\varepsilon}_{\mu\nu}{}^{\rho\sigma} = -\frac{1}{2}\left[\delta_\mu^\rho \delta_\nu^\rho \tilde{\nabla}^2 + \tilde{g}^{\rho\sigma} \tilde{\nabla}_\mu \tilde{\nabla}_\nu - \delta_\mu^\rho \tilde{\nabla}^\sigma \tilde{\nabla}_\nu - \delta_\nu^\rho \tilde{\nabla}^\sigma \tilde{\nabla}_\mu - \tilde{g}_{\mu\nu} \tilde{g}^{\rho\sigma} \tilde{\nabla}^2 + \tilde{g}_{\mu\nu} \tilde{\nabla}^\rho \tilde{\nabla}^\sigma \right]. \tag{2.16}$$

$\tilde{\nabla}_\mu$ is the covariant derivative compatible with the metric $\tilde{g}_{\mu\nu}$. Without loss of generality we can set $c = 1$. Equation 2.15 could be inverted so that a relation between the parameters β_n and the scale of the interaction m is expressed in terms of the parameters m_{FP} and α. In the following we will treat the parameters m_{FP} and α as independent.

For our purpose of modeling gravitational interaction in the solar system, a flat background on top of which gravitons propagate is needed. It is possible to find a combination of the parameters β_n such that Λ_g, and therefore Λ_f, are zero. Thus, we set $\tilde{g}_{\mu\nu} = \eta_{\mu\nu}$.

The action of the linearized BT with a flat background is then [19]

$$S_{LBT} = \frac{1}{2}\int d^4x\, G^{\mu\nu} \tilde{\varepsilon}_{\mu\nu}{}^{\rho\sigma} G_{\rho\sigma} + \int d^4x\, M^{\mu\nu} \left[\tilde{\varepsilon}_{\mu\nu}{}^{\rho\sigma} M_{\rho\sigma} + \frac{m_{FP}^2}{2} \left(M_{\mu\nu} - \tilde{g}_{\mu\nu} M \right) \right] \tag{2.17}$$

2.3 Vector Scalar Gravitational Interaction

To compute the time delay experienced by the propagation of a light signal in the Sun's gravitational field within a scattering-amplitude framework, the problem must be reformulated in the language of QFT. This requires specifying the matter fields participating in the interaction, as well as the mediators of the gravitational force. While the photon is naturally described by a vector field A_μ, and gravity is mediated by massless and massive spin-2 fields, the modeling of the Sun requires additional justification.

In the classical derivation of the Shapiro time delay in general relativity, the relevant quantity is the spacetime geometry in the exterior region of the source. At distances much larger than the radius of the Sun, the gravitational field is well approximated by its monopole contribution, and becomes insensitive to the internal structure and composition of the source. In this regime, the only parameter controlling the gravitational interaction is the total mass.

From the perspective of scattering amplitudes, this corresponds to coupling the gravitational field to an effective source characterized solely by its energy-momentum. Since gravity couples universally to the energy-momentum tensor, spin-dependent effects are subleading in the long-distance, weak-field limit considered here. Therefore, to leading order, the source can be modeled by the simplest field carrying mass, namely a massive scalar field.

This type of effective description is standard in amplitude-based approaches to classical gravity, where massive scalar fields are used to model heavy sources in processes such as gravitational scattering and radiation [20–22]. We therefore model the Sun as a massive scalar field ϕ with mass m_ϕ.

Another point worth mentioning is that, to preserve the ghost-free nature of BT, matter fields can only couple to one metric at a time [23, 24]. Thus, In this work we will require that both the photon field and the massive scalar field couple to the same metric $g_{\mu\nu}$.

The action of the scalar field is given by

$$S_\phi = -\frac{1}{2}\int d^4x\sqrt{-g}\left(\partial_\mu\phi\, g^{\mu\nu}\,\partial_\nu\phi + m_\phi^2\phi^2\right). \tag{2.18}$$

Using the perturbed form of the metric 2.10 and recalling that $\tilde{g}_{\mu\nu} = \eta_{\mu\nu} = (1+\alpha^2)\bar{g}_{\mu\nu}$, we get

$$g_{\mu\nu} = \frac{1}{1+\alpha^2}\eta_{\mu\nu} + \frac{1}{m_g}\delta g_{\mu\nu}. \tag{2.19}$$

To ensure canonical quantization, we need to rescale the field and its mass

$$\phi \to \frac{\phi}{\sqrt{1+\alpha^2}}, \qquad m_\phi \to \frac{m_\phi}{\sqrt{1+\alpha^2}}. \tag{2.20a}$$

Keeping only first order terms in the metric perturbation, the action of the rescaled field can be decomposed into a kinetic, $S_{K\phi}$ term and interaction term, $S_{I\phi}$.

$$S_{K\phi} = -\frac{1}{2}\int d^4x\left((\partial\phi)^2 + m_\phi^2\phi\right) \tag{2.21}$$

$$S_{I\phi} = -\frac{1}{2m_g(1+\alpha^2)}\int d^4x\delta g^{\mu\nu}\,\mathcal{T}_{\mu\nu}(\phi). \tag{2.22}$$

Where $T_{\mu\nu}$ is the energy momentum tensor of the scalar field given by

$$\mathcal{T}_{\mu\nu}(\phi) = \left[\partial_\mu\phi\partial_\nu\phi - \eta_{\mu\nu}\left((\partial\phi)^2 + m_\phi^2\phi^2\right)\right]. \tag{2.23}$$

In the previous equations, we have defined $(\partial\phi)^2 = \eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$. Inverting equations 2.11 and 2.12, we can express the interaction of the scalar field with the mass eigenstates $G_{\mu\nu}$ and $M_{\mu\nu}$.

$$\delta g^{\mu\nu} = (1+\alpha^2)\left(G^{\mu\nu} - 2\alpha^2 M^{\mu\nu}\right). \tag{2.24}$$

Hence, the interaction action is

$$S_{I\phi} = -\frac{1}{2m_g}\int d^4x \left(G^{\mu\nu} - 2\alpha^2 M^{\mu\nu}\right)\mathcal{T}_{\mu\nu}(\phi). \tag{2.25}$$

Consider the gravitational interaction of the photon field.

$$S_A = -\frac{1}{4}\int d^4x\sqrt{-g}F_{\mu\nu}g^{\mu\rho}g^{\nu\sigma}F_{\rho\sigma}. \tag{2.26}$$

This can also be decomposed into kinetic and interaction parts in the following manner

$$S_{k\phi} = -\frac{1}{4}\int d^4x F_{\mu\nu}F^{\mu\nu}, \tag{2.27a}$$

$$S_{IA} = \frac{1}{2m_g}\int d^4x \left(G^{\mu\nu} - 2\alpha^2 M^{\mu\nu}\right)\mathcal{T}_{\mu\nu}(A)\,. \tag{2.27b}$$

$\mathcal{T}_{\mu\nu}(A)$ is the energy momentum tensor of the photon field. It is given by

$$\mathcal{T}_{\mu\nu}(A) = F_{\rho\mu}F^{\rho}{}_{\nu} - \frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}\eta_{\mu\nu}\,. \tag{2.28}$$

At this stage it is possible to give the Feynman rules for the gravitational interaction of the Photon Field with a massive scalar. The photon propagator is given in the Feynman gauge while the massless graviton propagator is given in the de Deonder gauge [13].

- The propagator for the massive mode is

$$D^{(M)}_{\mu\nu,\rho\sigma}(p) = \frac{-i}{p^2 + m_{FP}^2 - i\epsilon}\left[\frac{1}{2}(P_{\mu\rho}P_{\nu\sigma} + P_{\mu\sigma}P_{\nu\rho}) - \frac{1}{3}P_{\mu\nu}P_{\rho\sigma}\right], \tag{2.29}$$

  with

$$P_{\mu\nu} = \eta_{\mu\nu} - \frac{p_\mu p_\nu}{m_{FP}^2}. \tag{2.30}$$

- The propagator for the massless mode is:

$$D^{(G)}_{\mu\nu,\rho\sigma}(p) = \frac{-i}{2(p^2 - i\epsilon)}\left(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho} - \eta_{\mu\nu}\eta_{\rho\sigma}\right). \tag{2.31}$$

- The propagator for the massive scalar is:

$$D^{(\phi)}(p) = \frac{-i}{p^2 + m_\phi^2 - i\epsilon}. \tag{2.32}$$

- The propagator of the photon is

$$D^{(A)}_{\alpha\beta}(k) = \frac{-i}{k^2 - i\epsilon}\eta_{\alpha\beta}. \tag{2.33}$$

- The massive scalar and massless graviton interaction vertex ($\gamma_G = \frac{1}{2m_g}$)

$$\tau^{G\phi}_{\mu\nu}(p,p') = i\gamma_G\left(p_\mu p'_\nu + p_\nu p'_\mu - \eta_{\mu\nu}(p\cdot p' - m_\phi^2)\right). \tag{2.34}$$

- The massive scalar and massive graviton interaction vertex ($\gamma_M = -2\alpha^2\gamma_G$)

$$\tau^{M\phi}_{\mu\nu}(p,p') = \frac{\gamma_M}{\gamma_G}\tau^{G\phi}_{\mu\nu}(p,p'). \tag{2.35}$$

- The massless graviton photon vertex (α and β are the photon polarization indices.)

$$\begin{aligned}\mathcal{T}^{\mu\nu;\alpha\beta}_{GA}(k_1,k_2) = 2i\,\gamma_G\Big[\,(k_1^\mu k_2^\nu + k_1^\nu k_2^\mu)\,\eta^{\alpha\beta} - k_1^\mu\eta^{\nu\beta}k_2^\alpha - k_2^\mu\eta^{\nu\alpha}k_1^\beta - k_1^\nu\eta^{\mu\beta}k_2^\alpha \\ - k_2^\nu\eta^{\mu\alpha}k_1^\beta + \eta^{\mu\nu}\left(k_1^\alpha k_2^\beta - (k_1\cdot k_2)\eta^{\alpha\beta}\right)\Big].\end{aligned} \tag{2.36a}$$

- The massive graviton photon vertex

$$\mathcal{T}^{\mu\nu;\alpha\beta}_{MA}(k_1,k_2) = \frac{\gamma_M}{\gamma_G}\mathcal{T}^{\mu\nu;\alpha\beta}_{GA}(k_1,k_2). \tag{2.37}$$

3 Eikonal Scattering Matrix and Time Delay Computation

The eikonal approximation is a high-energy semiclassical limit of quantum scattering in which the center-of-mass energy dominates over the momentum transfer, $s \gg |t|$. In this regime, the scattering is dominated by long-range interactions and corresponds classically to small-angle deflection, precisely the physical situation relevant to light propagation in the weak gravitational field of the Sun. A key feature of the eikonal approximation is that the scattering matrix exponentiates: the infinite series of ladder diagrams, which are leading at each order in the coupling, resums into a simple phase factor $e^{2i\chi}$, where χ is the eikonal phase [25, 26]. This exponentiation reduces the problem of computing an infinite class of Feynman diagrams to a single Fourier transform of the tree-level amplitude. The classical time delay is then extracted from the eikonal phase through differentiation with respect to the center-of-mass energy, via a stationary phase argument [11].

3.1 Kinematic Notations and Fourier Transforms Definitions

With the Feynman rules at hand, we can now tackle the central part of this paper. We consider the elastic gravitational scattering of an initial state of a scalar field with momentum p_1^μ and a photon field with momentum k_1^μ and polarization vector $\epsilon_1^{\lambda\mu}$. The final state also consists of a scalar field ϕ with momentum p_2^μ and a photon field with momentum k_2^μ and polarization λ_2. Let s be the center of mass energy $s = -(p_1+k_1)^2$ and q is the total momentum transfer $q = p_1 - p_2 = k_2 - k_1$. Regarding state normalization, we follow the conventions used in [13]:

$$2\pi\Theta(p_2^0)\delta(p_2^2+m_\phi^2)\,\langle p_2|p_1\rangle = (2\pi)^4\delta^4(p_2-p_1)\,, \tag{3.1a}$$
$$2\pi\Theta(k_2^0)\delta(k_2^2)\,\langle k_2,\lambda_2|k_1,\lambda_1\rangle = (2\pi)^4\delta^{\lambda_2}_{\lambda_1}\delta^4(k_2-k_1)\,. \tag{3.1b}$$

In the following we will suppose that the energies of all the particles in this work are positive and thus we drop the Θ function. Let S be the scattering operator,

$$\langle p_2,k_2,\lambda_2|S|p_1,k_1,\lambda_1\rangle = \langle p_2,k_2,\lambda_2|p_1,k_1,\lambda_1\rangle + i\,\langle p_2,k_2,\lambda_2|T|p_1,k_1,\lambda_1\rangle\,, \tag{3.2}$$

with T the transition matrix. We define the process amplitude by factorizing the energy momentum conservation from the transition matrix element :

$$\langle p_2, k_2, \lambda_2|T|p_1, k_1, \lambda_1\rangle = (2\pi)^4\delta^4\left(p_1 + k_1 - p_2 - k_2\right)\mathcal{M}(s, q, \lambda_1, \lambda_2). \tag{3.3}$$

Let $\mathcal{S}(s, q, \lambda_1, \lambda_2)$ be the function defined through

$$2\pi\delta(p_2^2 + m_\phi^2)2\pi\delta(k_2^2)\left\langle p_2, k_2, \lambda_2|S|p_1, k_1, \lambda_1\right\rangle = (2\pi)^4\delta^4\left(p_1 + k_1 - p_2 - k_2\right)\mathcal{S}(s, q, \lambda_1, \lambda_2). \tag{3.4}$$

.

$\mathcal{M}(s, q, \lambda_1, \lambda_2)$ and $\mathcal{S}(s, q, \lambda_1, \lambda_2)$ are related through

$$\mathcal{S}(s, q, \lambda_1, \lambda_2) = (2\pi)^4\delta_{\lambda_1}^{\lambda_2}\delta^4(q) + (2\pi)\delta(p_2^2 + m_\phi^2)(2\pi)\delta(k_2^2)\mathcal{M}(s, q, \lambda_1, \lambda_2), \tag{3.5a}$$

$$= (2\pi)^4\delta_{\lambda_1}^{\lambda_2}\delta^4(q) + (2\pi)\delta(-2p_1 \cdot q + q^2)(2\pi)\delta(2k_1 \cdot q + q^2)\mathcal{M}(s, q, \lambda_1, \lambda_2). \tag{3.5b}$$

The eikonal regime is when the momentum transfer is negligible compared to the center of mass energy. This is formalized by the following power counting:

$$s = -(p_1 + k_1)^2 >> -q^2 = t. \tag{3.6}$$

Thus, in this regime, equation 3.5 can be simplified to

$$\mathcal{S}(s, q, \lambda_1, \lambda_2) \approx (2\pi)^4\delta_{\lambda_1}^{\lambda_2}\delta^4(q) + (2\pi)\delta(-2p_1 \cdot q)(2\pi)\delta(2k_1 \cdot q)\mathcal{M}(s, q, \lambda_1, \lambda_2)$$

Finally, we define the Fourier transform to the impact parameter space of the scattering matrix element $\mathcal{S}(s, q, \lambda_1, \lambda_2)$

$$\tilde{\mathcal{S}}(s, b, \lambda_1, \lambda_2) = \int \frac{d^4q}{(2\pi)^4}\mathcal{S}(s, q, \lambda_1, \lambda_2), \tag{3.7a}$$

$$= \delta_{\lambda_1}^{\lambda_2} + \mathcal{FT}[\mathcal{M}](s, b, \lambda_1, \lambda_2), \tag{3.7b}$$

with

$$\mathcal{FT}[\mathcal{M}](s, b, \lambda_1, \lambda_2) = \int \frac{d^4q}{(2\pi)^4}(2\pi)\delta(-2p_1 \cdot q)(2\pi)\delta(2k_1 \cdot q)\mathcal{M}(s, q, \lambda_1, \lambda_2).$$

Equation 3.8 is the Fourier transform over the transverse momentum. While [13] provides a suitable frame to compute this transformation in the case of two massive or two massless incoming particles, this does not apply for the present case because the scalar field is massive and the photon field is not. Thus a new frame is needed. To be more precise we need two vectors x_1^μ and x_2^μ that satisfy the following conditions

1. x_1^μ and x_2^μ are linear combination of p_1^μ and k_1^μ.

2. Any 4-vector v^μ can be written as $v^\mu = v_1x_1^\mu + v_2x_2^\mu + v_\perp^\mu$, where $v_1 = 2p_1 \cdot v$, $v_2 = 2k_1 \cdot v$ and $v_\perp \cdot p_1 = v_\perp \cdot k_1 = 0$.

These conditions are satisfied by the following vector:

$$x_1^\mu = \frac{1}{(m_\phi^2 - s)} p_1^\mu + \frac{2m_\phi^2}{(m_\phi^2 - s)^2} k_1^\mu, \qquad x_2^\mu = \frac{1}{(m_\phi^2 - s)} k_1^\mu. \tag{3.8a}$$

The Jacobian of the transformation into these new variables is

$$d^4 q = \frac{1}{2(s - m_\phi^2)} dq_1 dq_2 d^2 q_\perp. \tag{3.9}$$

Thus, in the frame defined by the vectors x_1^μ and x_2^μ, the Fourier transform 3.8 takes the form

$$\mathcal{FT}[\mathcal{M}](s, b, \lambda_1, \lambda_2) = \frac{1}{2(s - m_\phi^2)} \int \frac{d^2 q_\perp}{(2\pi)^2} \mathcal{M}(s, q_\perp, \lambda_1, \lambda_2). \tag{3.10}$$

3.2 Tree Level Amplitudes and Vertices Simplifications

Because they play an important role in the final expression of the scattering matrix element and they also illustrate how the eikonal approximation simplifies computations, we propose to treat the tree level amplitudes in detail. Consider first the process where the photon field and scalar field scatter by the exchange of one massless graviton 1. Let ϵ^{λ_2} and ϵ^{λ_1} be the polarization vectors for the incoming and outgoing photon states. The amplitude for the process is given by

$$i\mathcal{M}_1^G(s, q, \lambda_1, \lambda_2) = \epsilon_\alpha^{\lambda_1} \epsilon_\beta^{\lambda_2 \dagger} \mathcal{T}_{GA}^{\mu\nu;\alpha\beta}(k_1, k_2) \mathcal{D}_{\mu\nu;\rho\sigma}^G(q) \tau_{G\phi}^{\rho\sigma}(p_1, p_2). \tag{3.11}$$

In the eikonal approximation, to leading order in the center of mass energy, the vertex

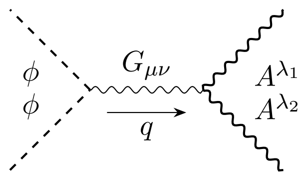

Figure 1: Tree level diagram for the gravitational scattering of ϕ and A^μ through the exchange of massless graviton.

coefficient simplifies to

$$\tau_{G\phi}^{\rho\sigma}(p_1, q - p_1) \approx -2i\gamma_g p_1^\rho p_1^\sigma. \tag{3.12}$$

The photon vertex also takes the form

$$\begin{aligned} \mathcal{T}_{GA}^{\mu\nu;\alpha\beta}(k_1, -k_1 - q) \approx -2i\,\gamma_G \Big[& 2k_1^\mu k_1^\nu \eta^{\alpha\beta} - k_1^\mu \eta^{\nu\beta} k_1^\alpha - k_1^\mu \eta^{\nu\alpha} k_1^\beta - k_1^\nu \eta^{\mu\beta} k_1^\alpha - k_1^\nu \eta^{\mu\alpha} k_1^\beta \\ & + \eta^{\mu\nu} k_1^\alpha k_1^\beta \Big]. \end{aligned} \tag{3.13a}$$

Thus, in the eikonal approximation to leading order in the center of mass energy, the tree level amplitude takes the simplified form

$$i\mathcal{M}_1^G(s, q, \lambda_1, \lambda_2) = \frac{2i\gamma_G^2}{q^2 - i\epsilon} \left(s - m_\phi^2\right)^2 \left(\epsilon^{\lambda_1} \cdot \epsilon^{\lambda_2 \dagger}\right), \tag{3.14a}$$

$$= \frac{2i\gamma_G^2}{q^2 - i\epsilon} \left(s - m_{\phi^2}\right)^2 \delta_{\lambda_1}^{\lambda_2}. \tag{3.14b}$$

Consider next the same tree level process mediated by the exchange of a massive graviton. In a similar manner to the massless graviton case, the amplitude for this process

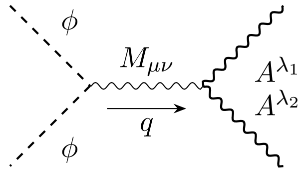

Figure 2: Tree level diagram for the gravitational scattering of ϕ and A^μ through the exchange of massive graviton.

is given by

$$i\mathcal{M}_1^M(s, q, \lambda_1, \lambda_2) = \frac{2i\gamma_M^2}{q^2 + m_{FP}^2} \left(s - m_{\phi^2}\right)^2 \delta_{\lambda_1}^{\lambda_2} \tag{3.15}$$

Notice that the only two changes compared to the massless case are the change of the coupling constant and the addition of the mass term in the denominator. To understand why this is the case, let $G(p, m)$ be the function

$$G(p, q) = \frac{1}{q^2 + m^2 - i\epsilon}. \tag{3.16}$$

The propagator of the massless and massive gravitons are,

$$\mathcal{D}^G_{\mu\nu;\rho\sigma}(q) = -iG(q, 0)L^G_{\mu\nu;\rho\sigma}\,, \tag{3.17a}$$
$$\mathcal{D}^M_{\mu\nu;\rho\sigma}(q) = -iG(q, m_{FP})L^M_{\mu\nu;\rho\sigma}(q)\,. \tag{3.17b}$$

$L^G_{\mu\nu;\rho\sigma}$ and $L^M_{\mu\nu;\rho\sigma}$ are given by

$$L^G_{\mu\nu;\rho\sigma} = \frac{1}{2}\left(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho} - \eta_{\mu\nu}\eta_{\rho\sigma}\right)\,, \tag{3.18a}$$
$$L^M_{\mu\nu;\rho\sigma} = \frac{1}{2}\left(Q_{\mu\rho}Q_{\nu\sigma} + Q_{\mu\sigma}Q_{\nu\rho}\right) - \frac{1}{3}Q_{\mu\nu}Q_{\rho\sigma}\,, \tag{3.18b}$$

with $Q_{\mu\nu} = \eta_{\mu\nu} - \frac{q_\mu q_\nu}{m_{FP}^2}$. In the eikonal regime the second term in $Q_{\mu\nu}$ is negligible because it is second order in the exchanged momenta. Thus, we get $Q_{\mu\nu} \approx \eta_{\mu\nu}$. This leads to

$$L^M_{\mu\nu;\rho\sigma} = L^G_{\mu\nu;\rho\sigma} + \frac{1}{6}\eta_{\mu\nu}\eta_{\rho\sigma}. \tag{3.19}$$

Then, because the photon vertex coefficient is traceless, $\eta_{\mu\nu}\mathcal{T}^{\mu\nu;\alpha\beta}_{GA}(k_1, -k_1 - q) = 0$, the only difference on the amplitude by the change of the massless propagator with the massive one is the addition of the mass in the denominator.

3.3 Eikonal Phase and Time Delay Computation

The general diagrams that we are interested in are those where at any given vertex only one graviton is emitted whether it is massive or massless. These diagrams are called *Ladder Diagrams*. Figure 3 shows an example of such diagrams with 4 exchanged gravitons. In

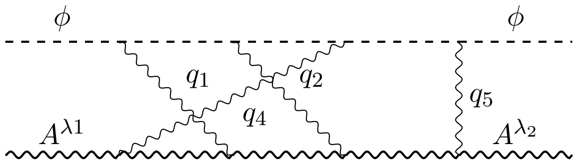

Figure 3: General ladder diagram due to the exchange of 4 gravitons.

the eikonal regime, for any given order in the expansion these diagrams are leading. To prove this we adapt the argument from [13]. Consider the two following cases depicted by figure 4: (i) the scalar fields emits two gravitons from two different vertices 4a. (ii) the two gravitons are emitted from the same vertex 4b. While the two cases are second order in the

(a) Emission from two different vertices.

(b) Emission from the same vertex.

Figure 4: Emission of two gravitons from different vertices (4a) and from the same vertex (4b).

coupling constant (γ_G or γ_M depending on the nature of gravitons), they differ in how they scale with the center of mass energy. In the first case, each vertex contributes a factor of $\gamma(s-m_\phi^2)$, and the scalar propagator introduces a factor of $\frac{1}{\sqrt{s-m_\phi^2}}$. The total contribution to the amplitude is therefore, $\gamma^2(s-m_\phi)^{\frac{2}{3}}$. In the second case, the contribution to the amplitude from the vertex is $\gamma^2(s-m_\phi^2)$ only. Thus a factor q will be needed to ensure the dimensional consistency. Thus the ladder diagrams are leading.

We now turn our focus to computing the amplitude of a general diagrams. The special feature of the eikonal approximation is that it allows for the expression of Fourier transform of the amplitude of the general diagram as an exponentiation of the Fourier transform of the tree level diagram. This is shown in details in [13, 27] for the case of linearized GR. See also [12, 28]. In appendix A we show that this also holds for BT,

$$\mathcal{FT}[\mathcal{M}_n](s,b,\lambda_1,\lambda_2) = \frac{1}{n!}\big(\mathcal{FT}[\mathcal{M}_1^G](s,q,\lambda_1,\lambda_2) + \mathcal{FT}[\mathcal{M}_1^M](s,q,\lambda_1,\lambda_2)\big)^n. \tag{3.20}$$

There are two observations that justify this result: (i) in first order linearized BT the massive and massless channels are decoupled and thus we expect there effects to add up.(ii) At the end of the previous subsection 3.2, we argued that the only difference between a process with a massless exchange or a massive one is in the change in the denominator of the propagator and the coupling constant. Thus, all projectors being the same, we expect only a multiplicative coefficient to change and hence for the same result for the massless case to hold for the mixed case.

The scattering matrix element in the impact parameter space is

$$\tilde{\mathcal{S}}(s,b,\lambda_1,\lambda_2) = \delta_{\lambda_1}^{\lambda_2} + \sum_{n=1}^{+\infty} \frac{1}{n!} \mathcal{FT}[\mathcal{M}]_n(s,b,\lambda_1,\lambda_2). \tag{3.21}$$

Averaging over initial polarization and summing over final ones we get

$$\tilde{\mathcal{S}}(s,b) = e^{2i\chi(s,b)}. \tag{3.22}$$

$\chi(s,b)$ is the eikonal phase and it is given by the polarization average of the Fourier transform of the tree level processes.

$$2\chi(s,b) = \frac{s-m_\phi^2}{4m_g^2} \int \frac{d^2q_\perp}{(2\pi)^2} \left(\frac{1}{q_\perp^2} + \frac{4\alpha^4}{q_\perp^2 + m_{FP}^2}\right) e^{ib\cdot q_\perp}. \tag{3.23}$$

Notice that the eikonal phase separate naturally into two contribution from massless mode and the massive without any interference between them. This is a result of the first order expansion of BT equations. The two gravity sectors act independently and at first order any phenomenological observation is expected to be the same of the two contributions. Furthermore, the massless contribution is that of GR [13], which is a well tested theory both on cosmological and astrophysical levels[29]. Thus, any deviation from it should be small, within the uncertainty levels of the observations. Hence, the massive mode contribution should then be suppressed. Equation 3.23 suggests two different ways to ensure that BT prediction stay fairly close to that of GR, either by taking α to zero or by taking m_{FP} to be large. One last observation regarding the computation of the eikonal phase, is that the integral in 3.23 is over all values of the momenta $q_\perp$ however for the eikonal power scale to be respected and hence the previous derivation to be consistent one needs to impose a cutoff Λ on the momentum amplitude. In this work, we take the cutoff to be the inverse of the smallest length characterizing the scattering: $\Lambda = \frac{1}{b_{min}}$, hence ensuring both the consistency with spacial resolution and also, in the case of light propagation in solar system, the eikonal power counting. The time delay is extracted from the eikonal phase, by first conducting a Fourier transform of the scattering matrix element to the time domain, then by applying the stationary phase approximation. This result in

$$\Delta T(s,b) = 2\frac{\partial \chi(s,b)}{\partial\sqrt{s}}\,, \tag{3.24a}$$

$$= \frac{\sqrt{s}}{2m_g^2} \int \frac{d^2q_\perp}{(2\pi)^2} \left(\frac{1}{q_\perp^2} + \frac{4\alpha^4}{q_\perp^2 + m_{FP}^2}\right) e^{ib\cdot q_\perp}\,. \tag{3.24b}$$

In polar coordinates this can be cast in the following form

$$\Delta T(s,b) = 4\sqrt{s}\,G \int_0^\Lambda J_0(bq_\perp) \left(\frac{1}{q_\perp} + \frac{4\alpha^4 q_\perp}{q_\perp^2 + m_{FP}^2}\right) dq_\perp. \tag{3.25}$$

J_0 is a first kind Bessel function,

$$J_0(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{x\cos(\theta)} d\theta, \tag{3.26}$$

and we have expressed the Planck mass m_g in terms of Newton gravitational constant $m_g^2 = \frac{1}{16\pi G}$. Note that in the massless mode contribution to the time delay,

$$\Delta T_G(s,b) = 4G\sqrt{s}\int_0^\Lambda J_0(bq_\perp)\frac{dq_\perp}{q_\perp}, \tag{3.27}$$

is infrared divergent. This is due to the long range nature of the force carried by GR's massless mode. In appendix B we provide an argument that the massless mode contribution reproduces the Shapiro time delay.

4 PN Parameter Computation and Parameter's Constraints From Solar System Measurements

In 1964, Shapiro proposed time delay measurements in the solar system as a fourth test for GR[8]. The test was conducted in 2002 by the Cassini spacecraft, a joint mission by NASA, ESA and the Italian Space Agency to study the planet Saturn. During a superior Solar conjunction, the spacecraft was on the other side of the sun with respect to earth. Radio signals were emitted from the ground station to the spacecraft and back. The roundtrip time delay was then measured and compared to GR prediction. The results were presented as constraints on the PPN parameter γ, which quantify how far a gravitational theory is from GR [9].

$$\gamma = 1 + (2.1 \pm 2.3)\times 10^{-5}. \tag{4.1}$$

In [10], the PPN parameter for linearized BT was obtained by solving the perturbed field equations to first order in γ for a point-like static source, and the resulting expression was used to constrain the parameter space of the theory. In this section, we first reproduce the dependence of the frequency shift on the impact parameter measured in the Cassini mission. We then extract the PPN parameter from the eikonal phase, use the Cassini constraints to restrict the parameter space of BT, and compare our results with those of [10].

4.1 Frequency Shift Computation

As discussed at the end of the previous section, the massless mode contribution to time delay is infrared divergent. Hence a better observable is the frequency shift of a photon due to gravitational interaction. This is, in fact, the exact observable measured by the Cassini mission. The frequency shift is linked to the time delay through

$$\mathcal{Y}(s,b) = \frac{\partial \Delta T}{\partial t}, \tag{4.2a}$$

$$= \frac{\partial b}{\partial t}\frac{\partial \Delta T}{\partial b}. \tag{4.2b}$$

In the following we will distinguish two contribution to the frequency shift $\mathcal{Y}_G$ and $\mathcal{Y}_M$ from the massless and massive channel respectively. Considering that the center of mass energy is equal to the mass of the Sun, M_s, we have

$$\mathcal{Y}_G(b) = 4iGM_s \frac{\partial b}{\partial t} \int_0^\Lambda J_1(bq_\perp) dq_\perp \,, \tag{4.3a}$$

$$\mathcal{Y}_M(b) = 16iG\alpha^4 M_s \frac{\partial b}{\partial t} \int_0^\Lambda \frac{J_1(bq_\perp) q_\perp^2}{q_\perp^2 + m_{FP}^2} dq_\perp \,. \tag{4.3b}$$

In the previous equations, the factor i cancel the imaginary contribution from the Bessel integral ensuring that the result is real. In [9], the authors argue that because the distance between the spacecraft and the Sun during the Solar conjunction (8.5 AU) was larger than that between earth and the Sun (1AU), the variation of the impact parameter with time could be approximated with the orbital velocity of earth v_e. Finally, for the ultraviolet cut-off we take the inverse of the smallest distance treated in the computation which is $b_{min} = 1.6R_\circ$, where $R_\circ$ is the Solar radius. Taking $t = 0$ the day of the solar conjunction, we get $b = v_e \cdot t + b_{min}$. Figure 5 depicts the gravitational frequency shift as a function of time, reproducing the measurements from Cassini data for $\alpha = 0.5$ and $m_{FP} = 10^{-40}$eV. The choice of this parameter was made so that we reproduce the correct qualitative curve and also to match the value of the Frequency shift at $t = 0$[1]. Reproducing the correct

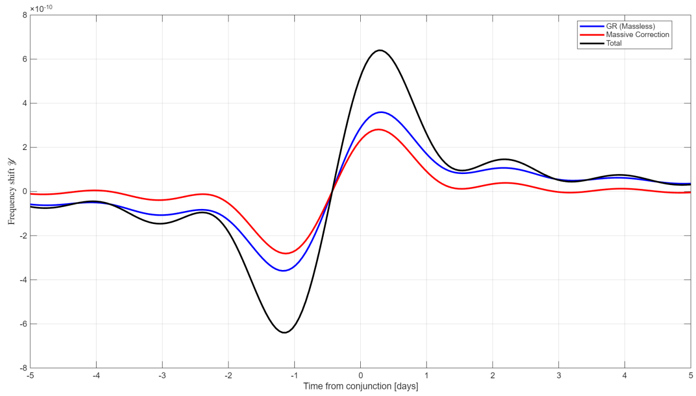

Figure 5: Frequency shift as a function of time for $\alpha = 0.5$, $m_{\rm FP} = 10^{-40}$ eV, and $\Lambda = \frac{1}{b_{\rm min}}$ eV.

qualitative and quantitative behavior as that measured by Cassini is a first check on the validity of the eikonal regime for this computation. It also shows that it is possible to find values for the parameters α and m_{FP} such that the deviation due to the massive mode is within the allowed region by the experimental results.

4.2 Post-Newtonian Parameter Derivation from Time Delay Measurements

For a gravitational theory characterized with a PPN parameter γ, the time delay is give by [9]

$$\Delta T(b) = (1 + \frac{\gamma - 1}{2}) \Delta T_G(b), \tag{4.4}$$

[1]The apparent oscillations are due the Fourier truncation, the period is on the order $T = \frac{2\pi}{\Lambda v_e}$, thus for bigger cut-off the oscillation are faster. It is possible to surpass this oscillation by introducing a filter during the integration i.e. by adding a term of the form $e^{(-q/\Lambda)^n}$ where n is an integer.

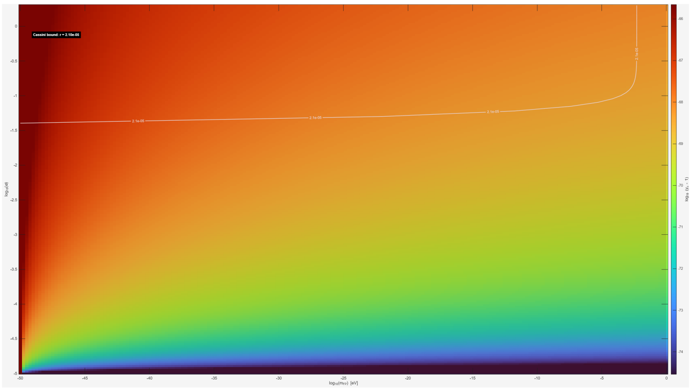

Figure 6: $\gamma_c - 1$ dependence on α and m_{FP} in log scale.

where $\Delta T_G(b)$ is the Shapiro time delay of GR. As shown in appendix B the massless contribution to the time delay reproduces the Shapiro time delay of GR. Thus, setting $\sqrt{s} = M_s$, we can rewrite equation 3.25 in the following form,

$$\Delta T(b) = (1 + \mathcal{Q}(b))\, \Delta T_G((b), \tag{4.5}$$

where $\mathcal{Q}$ is the ratio between the massive and massless contribution to the time delay. Comparing equations 4.4 and 4.5, we get

$$\gamma - 1 = 2\mathcal{Q}(b), \tag{4.6}$$

$$= 8\alpha^4 \frac{\left(\int_0^{b\Lambda} \frac{J_0(x)x}{x^2+(b\, m_{FP})^2} dx\right)}{\left(\int_0^{b\Lambda} \frac{J_0(x)}{x} dx\right)}. \tag{4.7}$$

In the previous equation we defined $x = bq_\perp$. The constraints for the Cassini mission are obtained for $b = b_{min}$, taking the cut-off $\Lambda = \frac{1}{b_{min}}$, we get

$$\gamma_c - 1 = 8\alpha^4 \frac{\left(\int_0^1 \frac{J_0(x)x}{x^2+(\frac{m_{FP}}{\Lambda})^2} dx\right)}{\left(\int_0^1 \frac{J_0(x)}{x} dx\right)}. \tag{4.8}$$

Figure 6, is a a log scale color map showing $\gamma_c - 1$ as a function of m_{FP} and α. When the mass is small, $\frac{m_{FP}}{\Lambda} \to 0$, equation 4.8 becomes $\gamma_c - 1 \approx 8\alpha^4$ thus the dependence of the constraints on m_{FP} decouples and the constraints is mainly on α. This explains the horizontal shape of the constraints for small masses. For small masses we get a value of $\alpha \approx 0.06$. For large masses, $\frac{m_{FP}}{\Lambda} >> 1$, $8\alpha^2 \frac{\Lambda^2}{m_{FP}^2}$ is an upper bound on $\gamma_c - 1$, Thus is the limit of large masses $\gamma_c - 1 \to 0$, hence the constraints is automatically satisfied.

The result obtained in [10] by solving the perturbed field equation of BT is

$$\gamma_f - 1 = \frac{-2\alpha^2 e^{-m_{FP} r}}{3 + 4\alpha^2 e^{-m_{FP} r}}. \tag{4.9}$$

In figure 7 we plot the relative difference between the eikonal and the field based calculation of the PPN parameter. We observe that for masses m_{FP} higher than 10^{-30} eV the difference between the two computation schemes is only relevant for impact paramters smaller than the Vainstein radius [30, 31]. However at this distance scale the linear approximation breaks [32, 33]. Thus in the linear model domain for masses higher than 10^{-30} eV the two calculation schemes are in agreement.

5 Conclusion

In this work we investigated the gravitational time delay in bimetric gravity using a scattering amplitude approach. Starting from the graviton–mediated interaction between a photon and a massive source, the corresponding scattering amplitude was computed and subsequently transformed to impact-parameter space in order to extract the eikonal phase. The phase shift encodes the classical propagation of the photon in the gravitational field of the source and allows the gravitational time delay to be obtained through differentiation with respect to the energy of the probe particle.

The resulting expression for the time delay contains the standard general relativistic contribution together with an additional term arising from the exchange of the massive spin-2 mode present in BT. This additional contribution modifies the effective PPN parameter (γ). The deviation from the general relativistic prediction depends on the coupling parameter controlling the interaction strength of the massive graviton and on the associated mass scale of the theory.

Using the observational bound on (γ) obtained from the Cassini spacecraft experiment, we translated the time delay expression into constraints on the parameters of the BT. The resulting bounds indicate that the coupling of the massive graviton to matter must remain sufficiently small in order to remain consistent with solar-system observations.

An interesting aspect of this analysis is that the classical gravitational observable is derived directly from the scattering amplitude via the eikonal phase. This illustrates how classical gravitational effects such as the Shapiro time delay can emerge from the underlying quantum field theoretic description of graviton exchange. In this sense, the eikonal formalism provides a useful bridge between perturbative scattering amplitudes and classical gravitational phenomena.

A possible extension of this work would be to consider higher-order expansions of BT around proportional background metrics with metric perturbations. Such an analysis would allow interaction terms between the massless and massive modes to appear, leading to additional contributions to the time-delay expression. This could provide a useful consistency test of BT, since contributions that reduce the total time delay could potentially lead to a negative net delay. The presence of such negative contributions would signal a violation of causality [34], thereby placing further constraints on the parameter space of the theory.

A Derivation of the General Ladder Diagram Contribution

In this appendix we derive equation 3.20. We first consider the case where all the exchanged gravitons are massless. This can be generalized to the mixed case easily by considering a

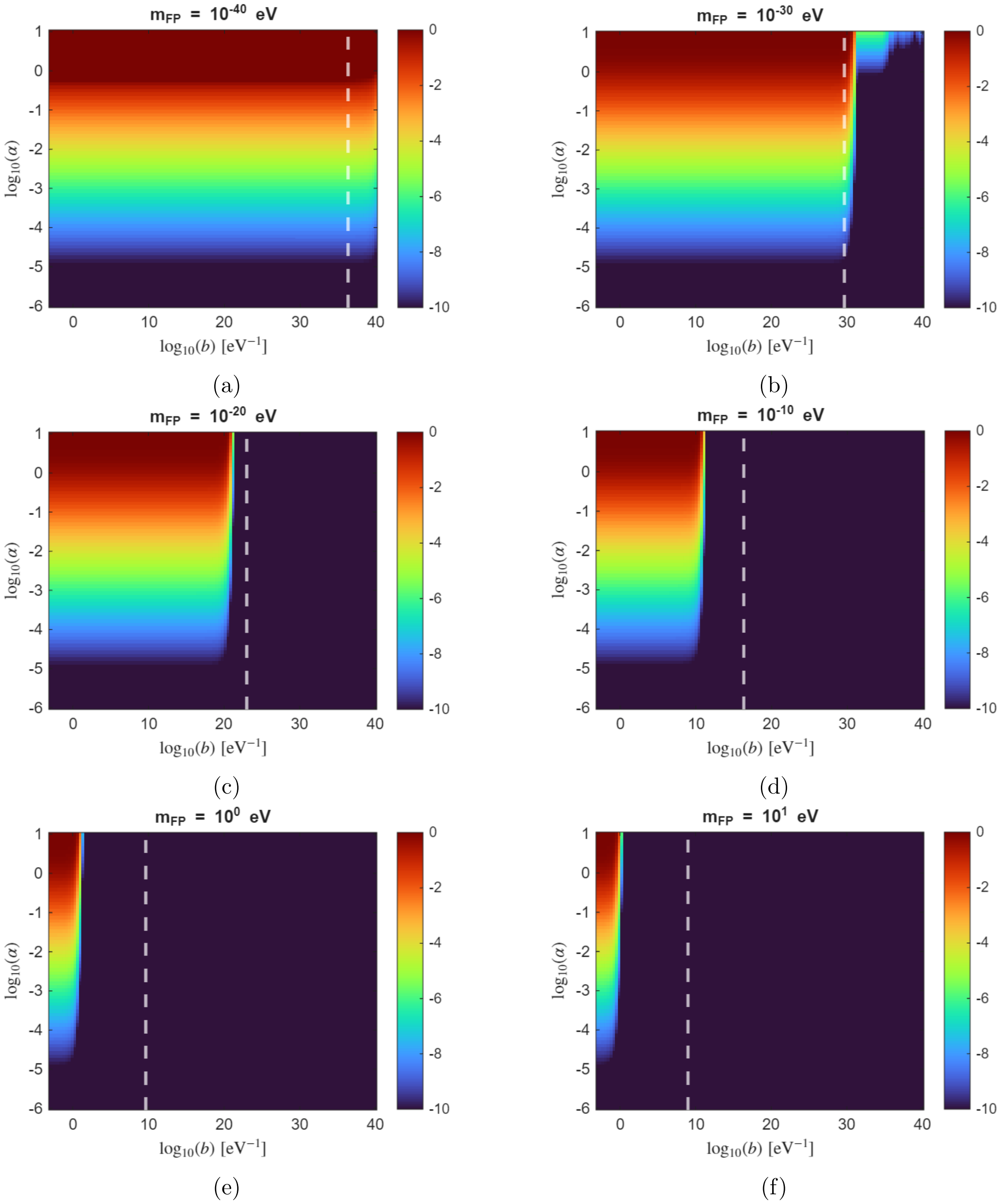

Figure 7: Relative difference between classical and eikonal PPN parameter γ for different Fierz–Pauli masses. The white dashed line indicates the Vainshtein radius, the distance after which the non linear effects are important and hence why the first order treatment fails. We observe that if the Fierz–Pauli mass is larger than 10^{-30} eV then the difference between the classical and eikonal calculation is limited for distances smaller than the Vainshtein radius.

sum over all possible combinations of massless and massive graviton exchanges that keep a total fixed number of vertices.

For a diagram with n exchanged gravitons, the scalar field line and the photon line each have n vertices, which are connected through the gravitons lines. We number these vertices and the exchanged momenta. A standard ladder diagram is then one where vertex i from the scalar line connects to vertex i of the photon line through the graviton momenta q_i. Figure 8 shows an example of such a diagram for $n = 4$. A general ladder diagram, with

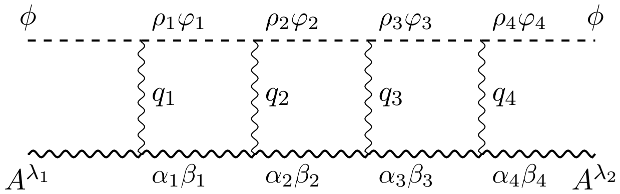

Figure 8: Standard Ladder with the exchange of 4 massless gravitons.

the same number of exchanged momenta, is obtain from the standard one by a permutation of the vertices that the momenta q_i connects. It is therefore specified by two permutation of n elements: σ_ϕ (σ_A) indicating which scalar (photon) vertex the momenta is connected to. For example, The diagram in Figure 9, is obtained with the permutation $\sigma_\phi = (1243)$ and $\sigma_A = (13)$.

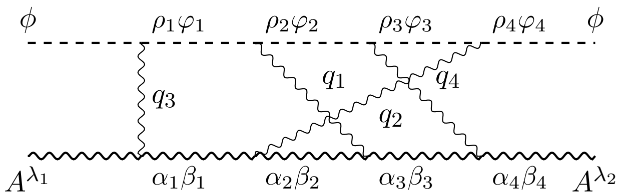

Figure 9: General Ladder diagram with the exchange of $n = 4$ gravitons and $\sigma_\phi = (1243)$ and $\sigma_A = (13)$.

The general diagram is constructed from the elementary units presented in figure 10. The contribution of such elementary unit is

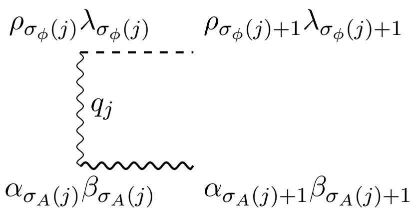

Figure 10: Elementary calculation unit.

$$\mathcal{D}^{G}_{\mu\nu;\rho_{\sigma_\phi(j)}\varphi_{\sigma_\phi(j)}}(q_j)\tau^{\rho_{\sigma_\phi(j)}\varphi_{\sigma_\phi(j)}}_{G\phi}\mathcal{T}^{\mu\nu;\alpha_{\sigma_A(j)}}_{GA\quad\alpha_{\sigma_A(j)+1}}G\left(p-\sum_{l=1}^{\sigma_\phi(j)}q_{\sigma_\phi^{-1}(l)},m_\phi\right)G\left(k+\sum_{l=1}^{\sigma_A(j)}q_{\sigma_A^{-1}(l)}\right). \tag{A.1}$$

The amplitude of the diagram is then obtained by multiplication of the correct n elementary units integrating over the momenta of the exchanged gravitons and multiplying by the in and out photons polarization vector. However this will result in an extra factor because the last elementary unit includes an extra propagators of the scalar field and the photon field. The correct amplitude is then

$$i\mathcal{M}^{\sigma_\phi,\sigma_A}_n(s,q,\lambda_1,\lambda_2)=\frac{(2\pi)^4\epsilon^{\lambda_1}_{\alpha}\epsilon^{\lambda_2\dagger}_{\beta}}{G(p_1-q,m_\phi)G(k_1+q,0)}\prod_{i=1}^{n}\int\frac{d^4q_j}{(2\pi)^4}\,\mathcal{D}^{G}_{\mu\nu;\rho_j\varphi_j}(q_j)\tau^{\rho_j\varphi_j}_{G\phi}\mathcal{T}^{\mu\nu;\alpha_j\beta_j}_{GA}$$
$$G\left(p-\sum_{l=1}^{j}q_{\sigma_\phi^{-1}(l)},m_\phi\right)G\left(k+\sum_{l=1}^{j}q_{\sigma_A^{-1}(l)},0\right)\delta^4\left((q-\sum_{j=1}^{n}q_j\right)\eta_{\beta_j\alpha_{j+1}}. \tag{A.2a}$$

In the previous equation we defined $\alpha_1=\alpha$ and $\beta_n=\beta$ and rearranged the terms. To get the total contribution of all such diagrams, we sum over all the permutations σ_ϕ and σ_A.However, because which the numbering of the momenta does not hold any physical information, we need to divide by $\frac{1}{n!}$ to avoid over counting. The case where both massless and massive graviton states contribute to the interaction is obtained by summing over all the different ways k out of n exchanged particles are chosen to be massless and the rest are massive. As we argued in 3.2, this will change $n-k$ coupling constant and graviton propagators to those of the massive mode with no other changes in the eikonal regime. Thus, the total contributions of all the different diagrams is given by [2]

$$i\mathcal{M}^{G}_n(s,q,\lambda_1,\lambda_2)=\frac{1}{n!}\sum_{\sigma_\phi,\sigma_A}i\mathcal{M}^{\sigma_\phi,\sigma_A}_n(s,q,\lambda_1,\lambda_2). \tag{A.3}$$

$$i\mathcal{M}^{n}(s,q,\lambda_1,\lambda_2)=\frac{1}{n!}\frac{(2\pi)^4\delta^{\lambda_2}_{\lambda_1}}{G(p_1-q,m_\phi)G(k_1+q,0)}\sum_{k=0}^{n}\mathcal{C}^n_k\left(2i\gamma_G(s-m_\phi^2)\right)^k\left(2i\gamma_G(s-m_\phi^2)\right)^{n-k}$$
$$\prod_{j=0}^{k}\int\frac{d^4q_j}{(2\pi)^4}\frac{1}{q_j^2-i\epsilon}\prod_{j=k+1}^{n}\int\frac{d^4q_l}{(2\pi)^4}\frac{1}{q_l^2+m_{FP}^2-i\epsilon}$$
$$\sum_{\sigma_\phi\sigma_A}G\big(p_1-\sum_{l=1}^{j}q_{\sigma_\phi(j)}\big)G\big(k_1+\sum_{l=1}^{j}q_{\sigma_A(j)}\big)\delta^4(q-\sum_{i=1}^{n}q_i).$$

At this stage the computation proceeds in an analogs manner to that in [13]. In the eikonal

[2] In this expression if the lower and upper arguments of the product simple are both zero than the product is 1 and if the lower argument is bigger than the upper one the product is one too.

regime, to lowest order in the center of mass energy we have

$$G\big(k_1 + \sum_{l=1}^{j} q_{\sigma_A(j)}\big) \approx \frac{1}{\sum_{l=1}^{n} 2k_1 \cdot q_{\sigma_A(j)}}, \tag{A.4a}$$

$$G\big(p_1 - \sum_{l=1}^{j} q_{\sigma_A(j)}\big) \approx \frac{1}{\sum_{l=1}^{n} -2p_1 \cdot q_{\sigma_A(j)}}. \tag{A.4b}$$

In the reference frame defined by the vector x_1, 3.8, and x_2, 3.8a. We have

$$\delta(q - \sum_{l=1}^{n} q_l) \approx 2(s - m_\phi)^2 \delta\left(\sum_{l=1}^{n} -2p_1 \cdot q_l\right) \delta\left(\sum_{l=1}^{n} 2k_1 \cdot q_l\right) \delta^2\left(q_\perp - \sum_{l=1}^{n} q_{l\perp}\right). \tag{A.5}$$

Then using the property

$$\delta(\sum_{l=1}^{n} w_n) \sum_\sigma \prod_{k=1}^{n} \frac{1}{\sum_{l=1}^{k} w_{\sigma_k}} = (2i\pi)^{n-1} \left(\sum_{i=1}^{n} w_i\right) \prod_{i=1}^{n} \delta(w_i), \tag{A.6}$$

the n gravitons amplitudes simplifies to by [3]

$$\begin{aligned} \frac{1}{2(s - m_\phi^2)} i\mathcal{M}^n(s, q_\perp, \lambda_1, \lambda_2) = \frac{1}{n} \delta_{\lambda_1}^{\lambda_2} \sum_{k=0}^{n} \mathcal{C}_k^n (2\pi)^2 \delta\left(q_\perp - \sum_{j=1}^{n} q_{j\perp}\right) \\ \prod_{j=0}^{k} \int \frac{d^4 q_j}{(2\pi)^4} 2\pi\delta(2p_1 \cdot q_j) 2\pi\delta(2k_1 \cdot q_j) \mathcal{M}_1^G(s, q_j) \\ \prod_{j=k+1}^{n} \frac{d^4 q_j}{(2\pi)^4} 2\pi\delta(2p_1 \cdot q_j) 2\pi\delta(2k_1 \cdot q_j) \mathcal{M}_1^M(s, q_j). \end{aligned} \tag{A.7}$$

Finally, transforming into the impact parameter space we get

$$\mathcal{FT}[\mathcal{M}^n](s, b, \lambda_1, \lambda_2) = \frac{1}{n!} \sum_{k=0}^{n} \mathcal{C}_k^n \left(\mathcal{FT}[\mathcal{M}_1^G](s, b, \lambda_1, \lambda_2)\right)^k \left(\mathcal{FT}[\mathcal{M}_1^M](s, b, \lambda_1, \lambda_2)\right)^{n-k}. \tag{A.8}$$

And hence, the Fourier transform to impact parameter space of any general ladder diagram is expressed in terms of those of the elementary tree level diagrams.

B Shapiro Time Delay from Massless mode contribution

The massless mode contribution to the time delay is given by equation 3.27.

$$\Delta T_G(s, b) = 4\sqrt{s} G \int_0^\Lambda J_0(bq_\perp) \frac{dq_\perp}{q_\perp}, \tag{B.1a}$$

$$= 4\sqrt{s} G \int_0^{b\Lambda} J_0(x) \frac{dx}{x}. \tag{B.1b}$$

[3] In this expression we omitted the dependence on the photon polarization index because the conservation of the polarization is already ensured by the $\delta(\lambda_1{}^{\lambda_2})$ term.

In the second line we changed the integration variable to $x = bq_\perp$. To compute the value of this integral we isolate the infrared divergent part.

$$\Delta T_G(s,b) = 4\sqrt{s}G \int_0^{b\Lambda} \frac{J_0(x)-1}{x} dx + 4\sqrt{s}G \int_0^{\Lambda} \frac{1}{x} dx. \tag{B.2}$$

Then using the property

$$\int_0^{b\Lambda} \frac{J_0(x)-1}{x} dx \xrightarrow{b\Lambda \to +\infty} -\ln\left(\frac{b\Lambda}{2}\right) - \gamma, \tag{B.3}$$

where γ is the Euler–Mascheroni constant, we get

$$\Delta T_G(s,b) = -4\sqrt{s}G \ln\left(\frac{b\Lambda}{2}\right) + C + IR, \tag{B.4}$$

with IR is the infrared divergence and c is a constant independent of b. keeping only the terms dependent on b, we get

$$\Delta T_G(s,b) = 2\sqrt{s}G \ln \frac{4}{b^2\Lambda^2}. \tag{B.5}$$

If we consider that the center of mass energy is dominated by the mass of the source of the gravitational field, which is the case for light propagation in the Sun's vicinity,we get $(\mu = \frac{2}{\Lambda})$

$$\Delta T_G(s,b) = 2M_s G \ln\left(\frac{\mu^2}{b^2}\right), \tag{B.6}$$

with M_s the mass of the Sun. The Shapiro time delay is given by

$$\Delta T_S = 2M_s G \ln(\left(\frac{4r_e r_s}{b^2}\right), \tag{B.7}$$

with r_e and r_s the distance between the source and the emitter and receiver of the light signal respectively. Thus if we take $\Lambda = \frac{1}{\sqrt{r_e r_s}}$ the massless contribution to the time delay reproduces the Shapiro result.

Acknowledgments

The author would like to thank Dr. Fawad Hassan for his support and supervision during the derivation and writing of the paper.

References

[1] D. Boulware and S. Desser, *Inconsistency of finite range gravitation*, *Physics Letters B* **40** (1972) 227.

[2] M.S. Volkov, *Exact self-accelerating cosmologies in ghost-free bigravity and massive gravity*, *Phys. Rev. D* **86** (2012) 061502.

[3] K. Aoki and K.-i. Maeda, *Cosmology in ghost-free bigravity theory with twin matter fluids: The origin of dark matter*, *Phys. Rev. D* **89** (2014) 064051.

[4] M. Lüben, A. Schmidt-May and J. Weller, *Physical parameter space of bimetric theory and sn1a constraints*, *Journal of Cosmology and Astroparticle Physics* **2020** (2020) 024.

[5] F. Koennig, Y. Akrami, L. Amendola, M. Motta and A.R. Solomon, *Stable and unstable cosmological models in bimetric massive gravity*, *Phys. Rev. D* **90** (2014) 124014.

[6] K. Max, M. Platscher and J. Smirnov, *Gravitational wave oscillations in bigravity*, *Physical Review D* **97** (2018) 064009.

[7] M. Fasiello and R.H. Ribeiro, *Mild bounds on bigravity from primordial gravitational waves*, *Journal of Cosmology and Astroparticle Physics* **2015** (2015) 027.

[8] I.I. Shapiro, *Fourth test of general relativity*, *Phys. Rev. Lett.* **13** (1964) 789.

[9] B. Bertotti, L. Iess and P. Tortora, *A test of general relativity using radio links with the cassini spacecraft*, *Nature* **425** (2003) 374.

[10] M. Hohmann, *Post-newtonian parameter γ and the deflection of light in ghost-free massive bimetric gravity*, *Phys. Rev. D* **95** (2017) 124049.

[11] Y. Ma and D.-f. Zeng, *Gravitational deflection of vector photons via effective field theory*, *Phys. Rev. D* **113** (2026) 016017.

[12] G. 't Hooft, *Graviton dominance in ultra-high-energy scattering*, *Physics Letters B* **198** (1987) 61.

[13] P. Di Vecchia, C. Heissenberg, R. Russo and G. Veneziano, *The gravitational eikonal: From particle, string and brane collisions to black-hole encounters*, *Physics Reports* **1083** (2024) 1.

[14] M. Accettulli Huber, A. Brandhuber, S. De Angelis and G. Travaglini, *Eikonal phase matrix, deflection angle, and time delay in effective field theories of gravity*, *Phys. Rev. D* **102** (2020) 046014.

[15] S.F. Hassan and R.A. Rosen, *Resolving the ghost problem in nonlinear massive gravity*, *Phys. Rev. Lett.* **108** (2012) 041101.

[16] A. Schmidt-May and M.v. Strauss, *Recent developments in bimetric theory*, *Journal of Physics A: Mathematical and Theoretical* **49** (2016) 183001.

[17] S.F. Hassan, A. Schmidt-May and M. von Strauss, *Particular solutions in bimetric theory and their implications*, *International Journal of Modern Physics D* **23** (2014) 1443002 [https://doi.org/10.1142/S0218271814430020].

[18] M. Fierz and W.E. Pauli, *On relativistic wave equations for particles of arbitrary spin in an electromagnetic field*, *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* **173** (1939) 211 [https://royalsocietypublishing.org/rspa/article-pdf/173/953/211/37464/rspa.1939.0140.pdf].

[19] S.F. Hassan, A. Schmidt-May and M.v. Strauss, *Bimetric theory and partial masslessness with lanczos–lovelock terms in arbitrary dimensions*, *Classical and Quantum Gravity* **30** (2013) 184010.

[20] C. Cheung, I.Z. Rothstein and M.P. Solon, *From scattering amplitudes to classical potentials in the post-minkowskian expansion*, *Phys. Rev. Lett.* **121** (2018) 251101.

[21] N.E.J. Bjerrum-Bohr, P.H. Damgaard, L. Planté and P. Vanhove, *The sagex review on scattering amplitudes chapter 13: Post-minkowskian expansion from scattering amplitudes*, *Journal of Physics A: Mathematical and Theoretical* **55** (2022) 443014.

[22] A. Buonanno and T. Damour, *Effective one-body approach to general relativistic two-body dynamics*, Phys. Rev. D **59** (1999) 084006 [gr-qc/9811091].

[23] C.d. Rham, L. Heisenberg and R.H. Ribeiro, *On couplings to matter in massive (bi-)gravity*, *Classical and Quantum Gravity* **32** (2015) 035022.

[24] Y. Yamashita, A. De Felice and T. Tanaka, *Appearance of boulware–deser ghost in bigravity with doubly coupled matter*, *International Journal of Modern Physics D* **23** (2014) 1443003 [https://doi.org/10.1142/S0218271814430032].

[25] M. Lévy and J. Sucher, *Eikonal approximation in quantum field theory*, *Phys. Rev.* **186** (1969) 1656.

[26] D. Kabat and M. Ortiz, *Eikonal quantum gravity and planckian scattering*, *Nuclear Physics B* **388** (1992) 570.

[27] D. Amati, M. Ciafaloni and G. Veneziano, *Superstring collisions at planckian energies*, *Physics Letters B* **197** (1987) 81.

[28] I.J. Muzinich and S. Vokos, *Long range forces in quantum gravity*, *Phys. Rev. D* **52** (1995) 3472.

[29] C.M. Will, *The confrontation between general relativity and experiment*, *Living Reviews in Relativity* **17** (2014) 4.

[30] C. de Rham, L. Heisenberg and R.H. Ribeiro, *On the vainshtein mechanism in the minimal model of massive gravity*, *Physical Review D* **91** (2015) 024018.

[31] A. Vainshtein, *To the problem of nonvanishing gravitation mass*, *Physics Letters B* **39** (1972) 393.

[32] E. Babichev and M. Crisostomi, *Restoring general relativity in massive bigravity theory*, *Physical Review D* **88** (2013) 084002 [1307.3640].

[33] G. Chkareuli and D. Pirtskhalava, *Vainshtein mechanism in λ_3-theories*, *Physics Letters B* **713** (2012) 99 [1105.1783].

[34] X.O. Camanho, J.D. Edelstein, J. Maldacena and A. Zhiboedov, *Causality constraints on corrections to the graviton three-point coupling*, *Journal of High Energy Physics* **2016** (2016) 20.